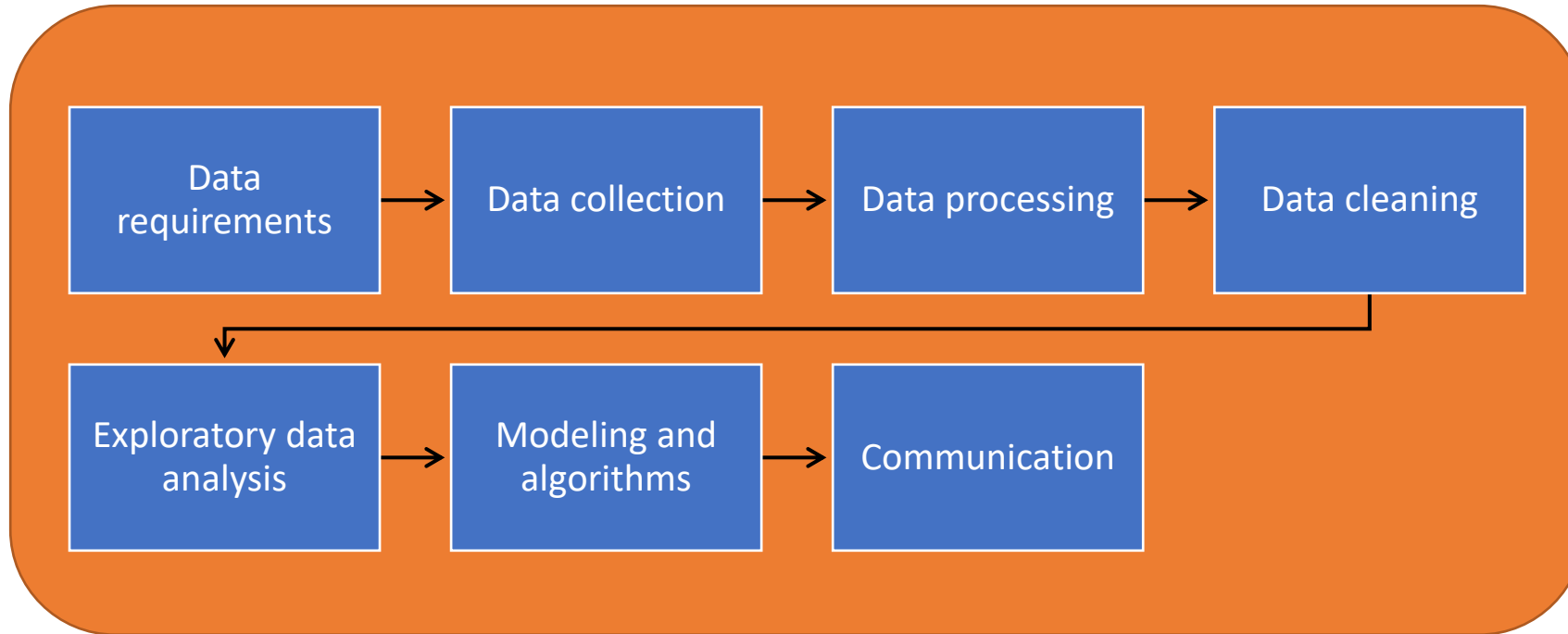


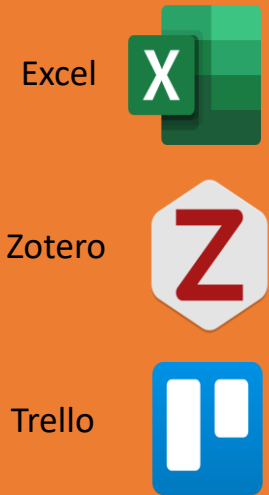
Types of Data Analytics



Process: Data Analysis



Tools



Techniques



Descriptive Analytics
Diagnostic Analytics
Predictive Analytics
Prescriptive Analytics

Methods



Considerations



Learning Outcomes for Types of Data Analytics

We will discuss the 4 types of Data Analytics:

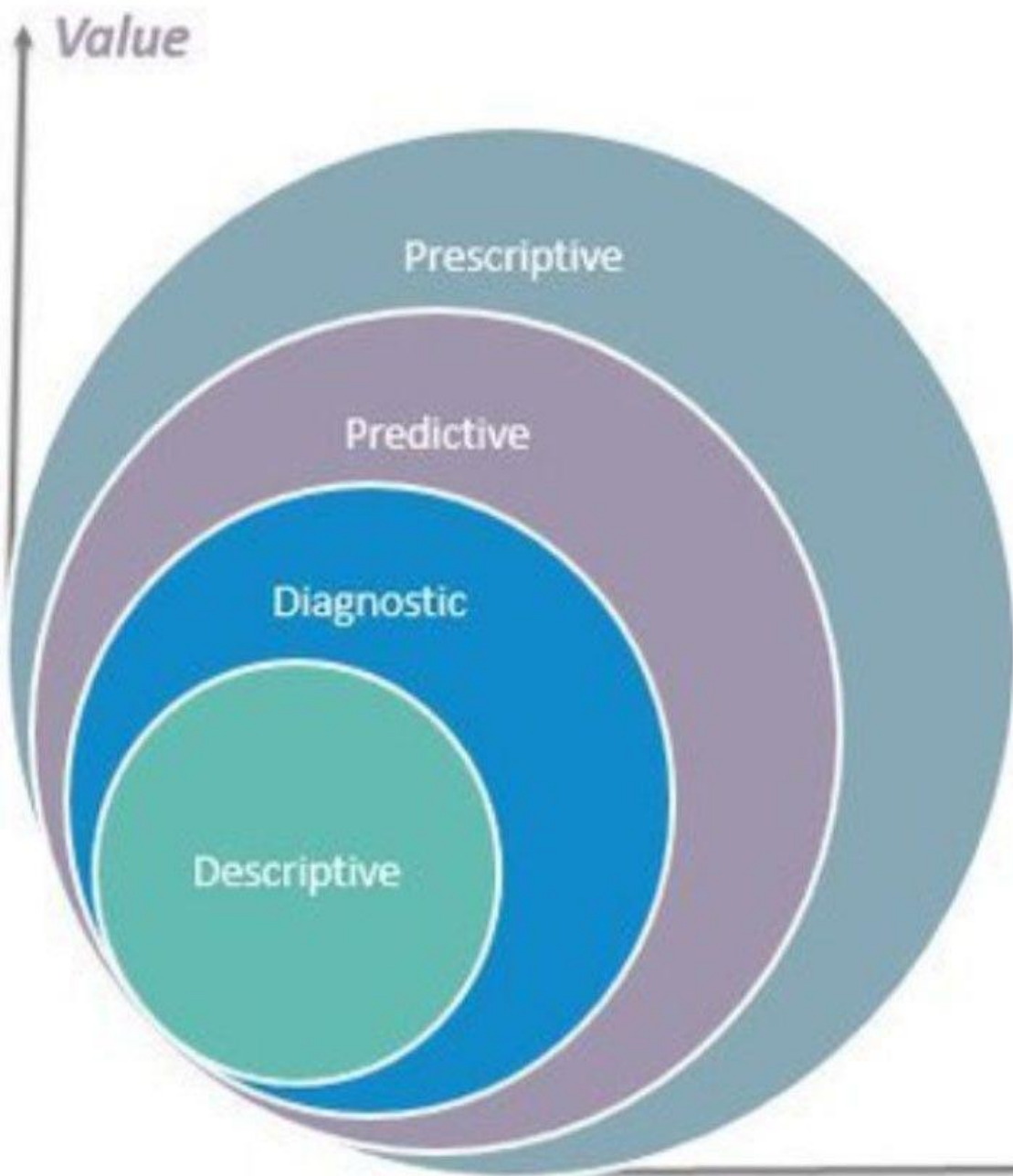
- Descriptive Analytics.
- Diagnostic Analytics.
- Predictive Analytics.
- Prescriptive Analytics.

Introduction

- Data **analysis** can be classified into **four types or categories**, depending on the **question to which the analysis answers**. These four types of data analytics **can help** organization strategists and decision makers:
 1. Describe previous results (Descriptive Analytics).
 2. Diagnose the reason for these results (Diagnostic Analytics).
 3. Predict what might happen in the future (Predictive Analytics).
 4. Recommend actions to the organization for the next steps (Prescriptive Analytics).



4 types of Data Analytics



What is the data telling you?

Descriptive: *What's happening in my business?*

- Comprehensive, accurate and live data
- Effective visualisation

Diagnostic: *Why is it happening?*

- Ability to drill down to the root-cause
- Ability to isolate all confounding information

Predictive: *What's likely to happen?*

- Business strategies have remained fairly consistent over time
- Historical patterns being used to predict specific outcomes using algorithms
- Decisions are automated using algorithms and technology

Prescriptive: *What do I need to do?*

- Recommended actions and strategies based on champion / challenger testing strategy outcomes
- Applying advanced analytical techniques to make specific recommendations

Complexity

Types of Data Analytics Table of Contents

1. Descriptive Analytics 
2. Diagnostic Analytics
3. Predictive Analytics
4. Prescriptive Analytics

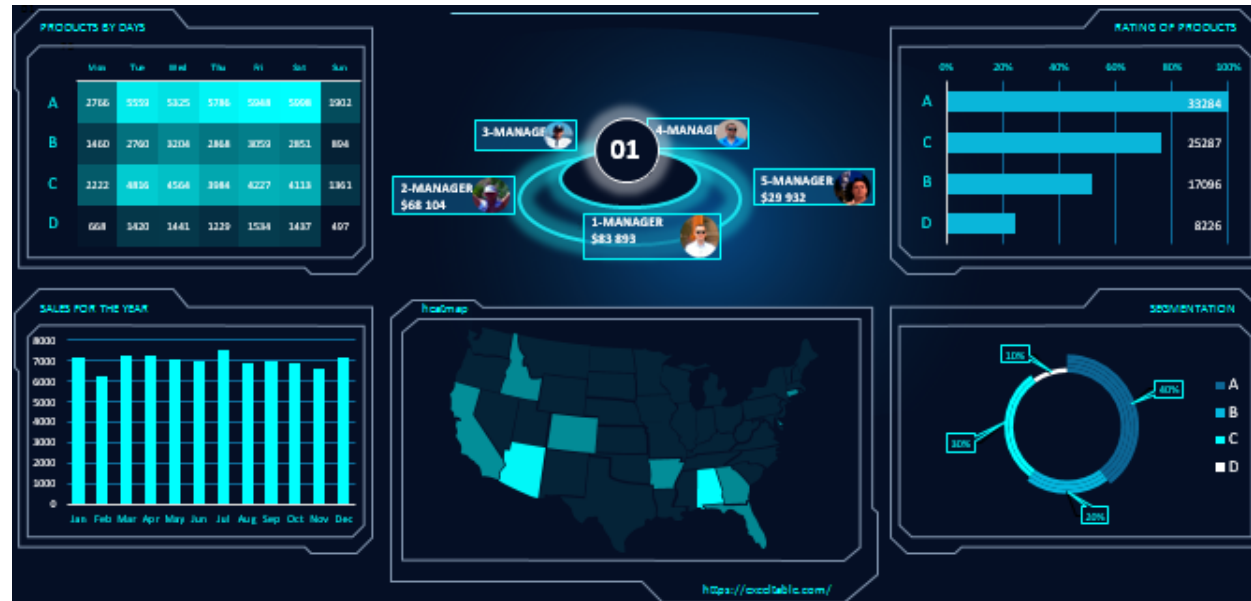
Descriptive Analytics

- “**What Happened**”
- This is the **most common** of all forms.
- **Focus** on the **Past**.
- The purpose of descriptive analytics is simply to **summarize and tell you what happened**.



Descriptive Analytics

- **Describes or summarizes the available data.**
- Static or dynamic reports that summarize measures, such as performance metrics, and **provide the foundation** for most **business intelligence (BI) tools and dashboards.**



Descriptive Analytics

- Descriptive Analytics can serve as a **starting point** or basis for creating an **analysis strategy**, as some professionals estimate that 80% of business analyses consist of **descriptions made from aggregations of previous results**.



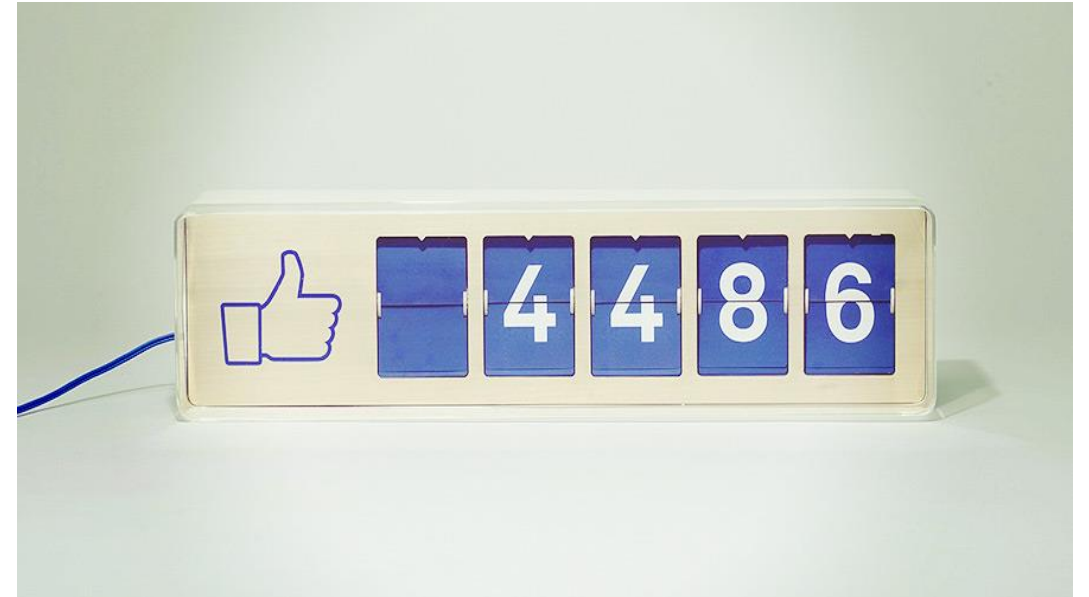
Descriptive Analytics

- It is an important step for investors, shareholders and managers to **understand raw data**.
- This facilitates the **identification** and treatment of **strengths** and **weaknesses**, which can **help the preparation of strategies**.



Descriptive Analytics

- For example, number of post, mentions, fans, followers, page views, check-ins, pins, etc. (simple event counter)
- Other descriptive analytics may be results of **simple arithmetic operations**, such as share of voice, average response time, % index, average number of replies per post, etc.



Descriptive Analytics

- Techniques like **data aggregation**, data mining, clustering and/or summary statistics provide analytics that **describe** a **past state** and can be used for **descriptive analytics**.
- The most commonly used technique is simple descriptive statistics and data visualization in graphical form.



Hospital example (I)

- Let's assume that in our **hospital** an **unusually high number of people** are admitted to the **emergency room** in a **short period of time**.
- **Descriptive analytics** tells you **that this is happening** and **provides real-time data** with all the corresponding statistics (date of occurrence, volume, patient details, etc.).



<https://public.tableau.com/app/profile/abdelaziz.mahjoub/viz/RWFDHospitalEmergencyroomvisits/RWFDHospitalsEmergencyroom>

Hospital example (I)

- What data would we need to know that an **unusually** high number of people are admitted to the **emergency room** in a **short period** of time?
- How could these data be obtained?
- How often should they be updated?
- To whom would such information be provided, how often and in what format?



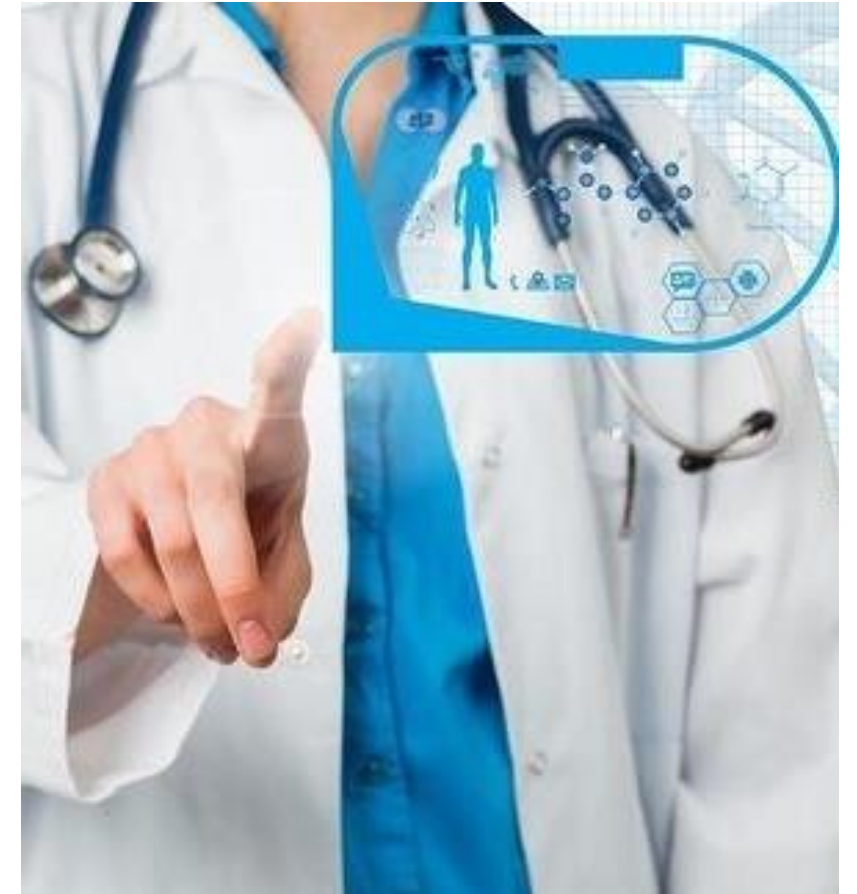
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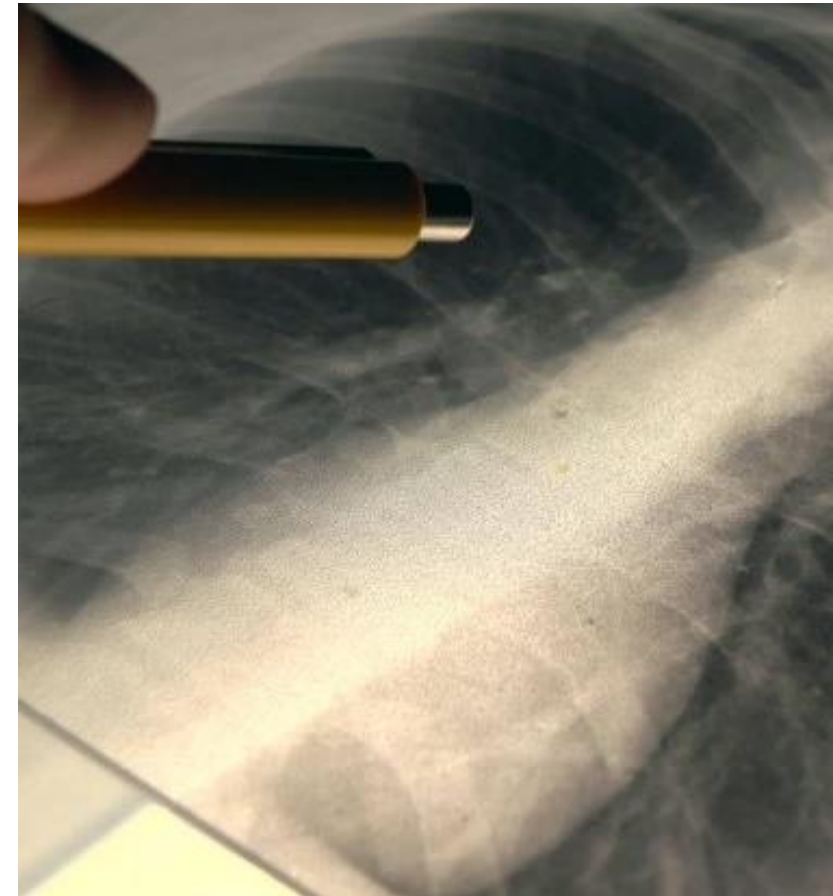
Diagnostic Analytics

- “Why Did it Happen”
- **Builds upon** Descriptive Analytics.
- Like descriptive analytics, diagnostic analytics also **focus** on the **past**.



Diagnostic Analytics

- Diagnostic Analytics **look** for **cause** and **effect** to illustrate **why** something **occurred**.
- The **objective** is to **compare past occurrences** to **determine causes**.
- Of course, this is **seldom** a **black-and-white** answer, but instead provided in the context of **probability, likelihood** or a **distributed outcome**.



Diagnostic Analytics

Diagnostic analytics can provide guidance by helping to:

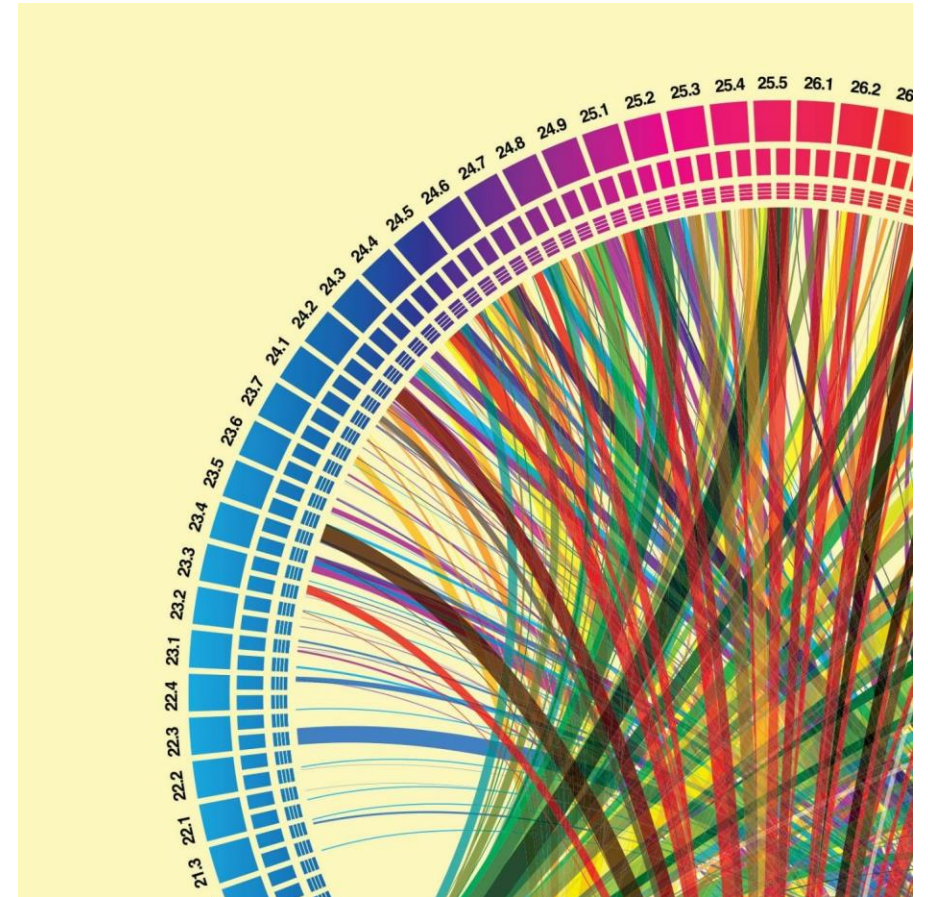
- **Identify outliers:** For example, a sudden drop in sales or an explosion in website traffic that can't be explained may indicate a need for additional examination.



Diagnostic Analytics

Diagnostic analytics can provide guidance by helping to:

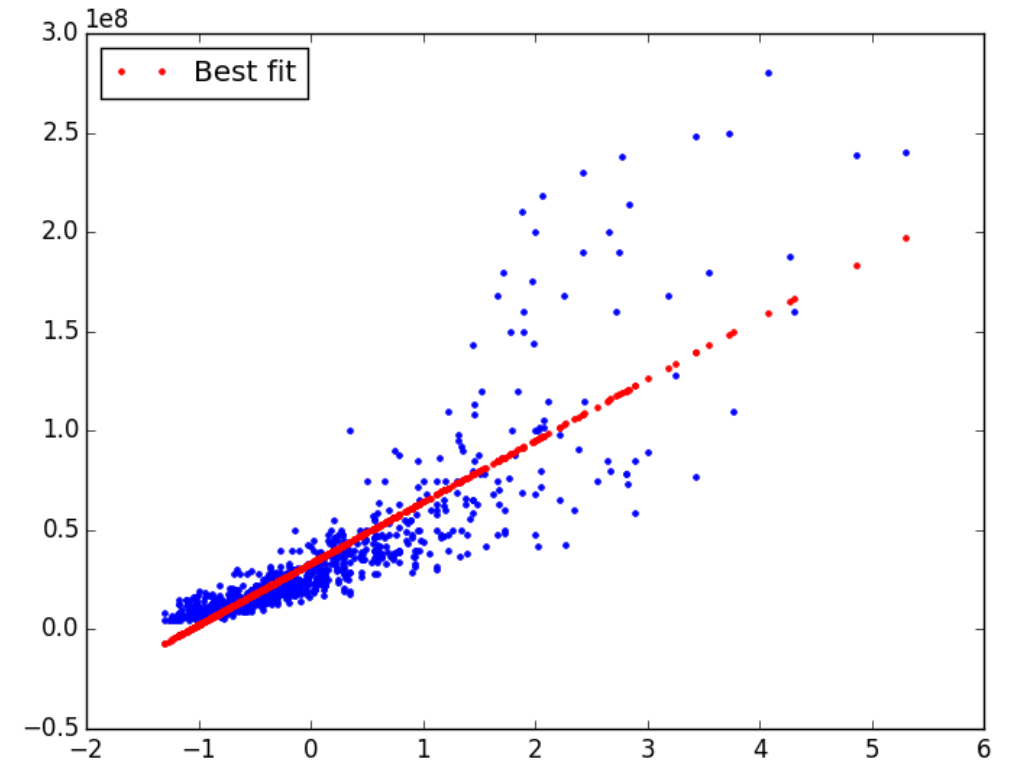
- **Isolate patterns:** Analysts may need to look outside the existing dataset to identify the source of the pattern. For example, a sudden drop in sales may have stemmed from the launch of a disruptor.



Diagnostic Analytics

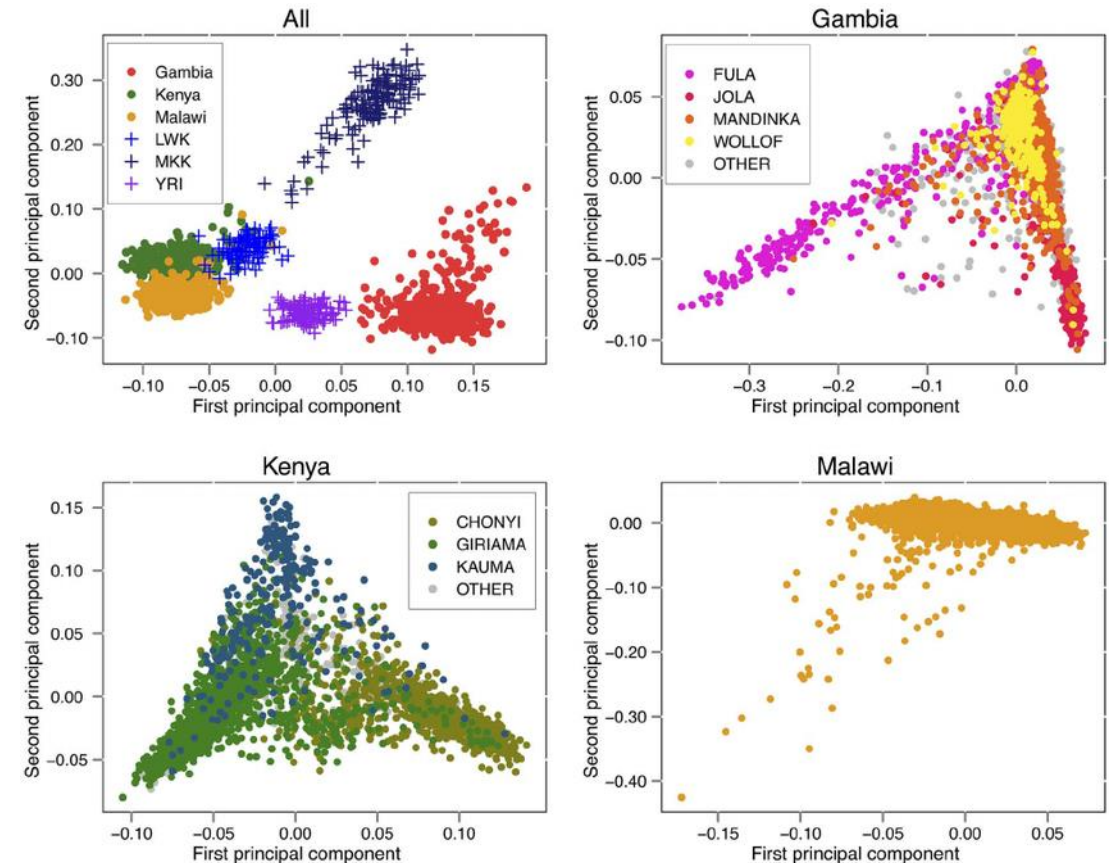
Diagnostic analytics can provide guidance by helping to:

- **Uncover relationships:** Using more complex analytics, analysts may employ **probability** theory, **regression** analysis, or time series to isolate **cause** and **effect** relationships.



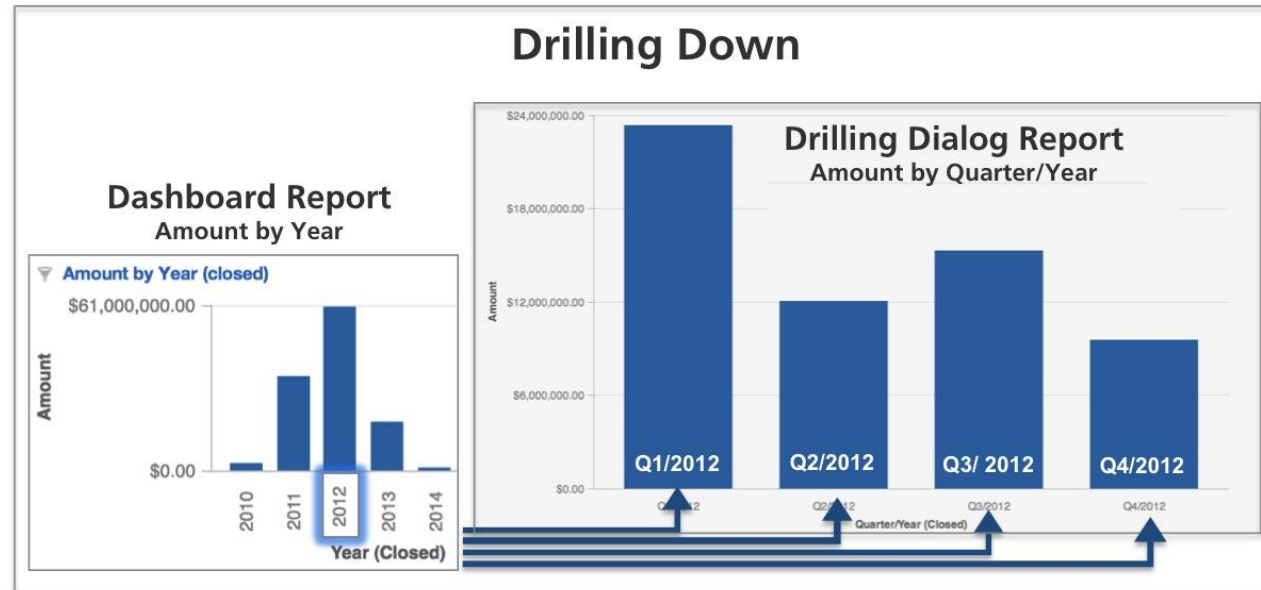
Diagnostic Analytics

- To determine a “why” from your data, applying such techniques as Principal Components Analysis, Sensitivity Analysis and Training Algorithms for Classification and Regression Analysis can all provide a diagnosis—diagnostic analytics.



Diagnostic Analytics

- Using descriptive data, diagnostic analytical tools will empower an analyst to drill down and in so doing isolate the root cause of a problem.



Hospital example (II)

- **Diagnostic analytics** would **explore** the **data** and **make correlations**. For instance, it may help you determine that all of the patients' **symptoms**—high fever, dry cough, and fatigue—point to the same infectious agent.
- You now have an explanation for the sudden spike in volume at the ER.



Types of Data Analytics Table of Contents

1. Descriptive Analytics

2. Diagnostic Analytics

3. Predictive Analytics



4. Prescriptive Analytics

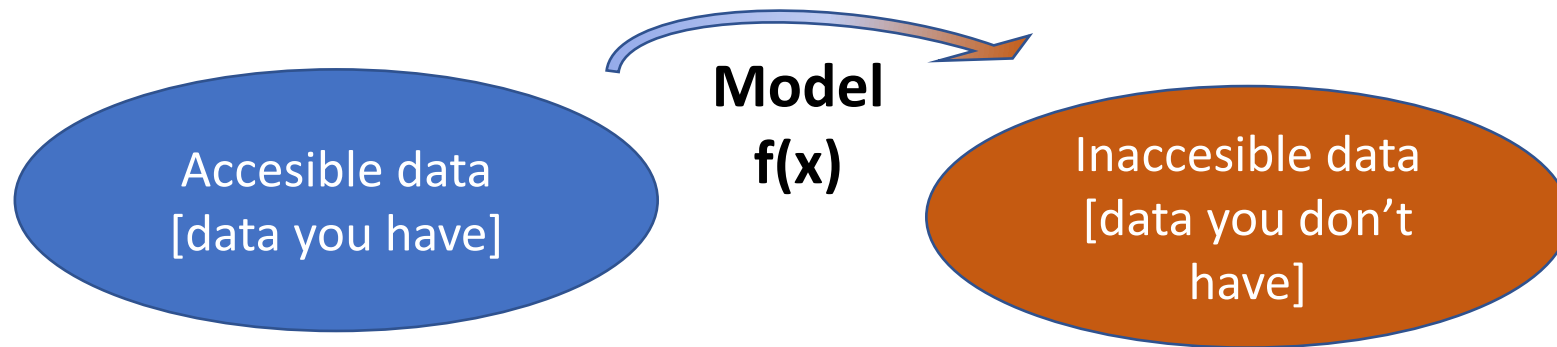
Predictive Analytics

- Predictive analytics is all about **forecasting**.
- With predictive analytics, a data analyst uses **quantitative analysis** of the dataset and establishes **predictions**, often called **predictive modeling**.



Predictive Analytics

Predictive analytics in a picture



Predictive Analytics

- In a world of significant uncertainty, being able to predict allows one to make **better decisions**.
- Predictive models are some of the most important ones and are used across many fields.



Predictive Analytics

- The **purpose** of predictive analytics is **NOT to tell you what will happen in the future**. It cannot do that. In fact, no analytics can do that.
- Predictive **analytics can only forecast what might happen in the future**, because all predictive analytics are **probabilistic** in nature.



Predictive Analytics

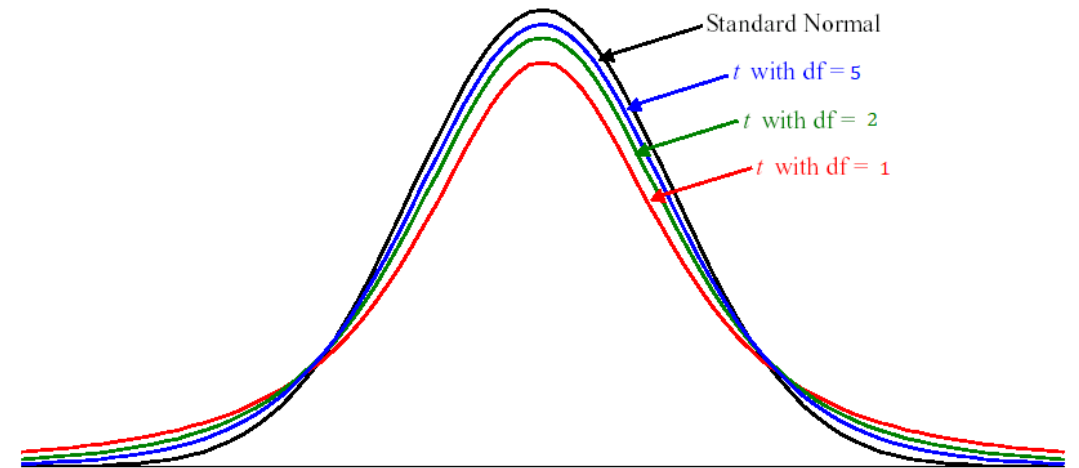
- Predictive models typically use a range of variables to make predictions.
- The variability of these input variables is often related to the predicted outcome.
- For example, as a person ages, the likelihood of a heart attack increases, indicating a positive correlation between age and heart attack risk. In this case, age serves as a predictor with an increasing relationship to heart attack risk.



Predictive Analytics

- Confidence intervals, T statistics, K-S statistics, and P values are all part of the umbrella of predictive analytics, applied in Machine Learning algorithms, Classification Models and Regression Models, to name some examples.

Student's t -distribution



Predictive Analytics

- Building these models **requires expertise**, typically from data scientists, machine learning experts or other data analytics specialists working under a data analytics manager.

The screenshot displays the CUNEF website's navigation and program details. At the top, the CUNEF logo is on the left, and a navigation bar includes links for Library, Admissions, Alumni, Contact, Faculty & Staff, and flags for the UK and Spain. A 'STUDENTS ACCESS' button is highlighted. Below this is a secondary menu with links for CUNEF, STUDIES, FACULTY & RESEARCH, INTERNATIONAL, CLUBS & SPORTS, CAREERS & EMPLOYMENT, and NEWS. The breadcrumb trail reads 'CUNEF > ESTUDIOS > CUNEF GRADUATE'. Three tabs are visible: 'CUNEF UNDERGRADUATE', 'CUNEF GRADUATE' (which is selected), and 'CUNEF EXECUTIVE'. The main content area features a large image of students in a classroom with the text 'Master in Data Science for Finance' overlaid. At the bottom, program details are listed: PROGRAM: Postgraduate courses, MODE: Full-time, LANGUAGE: Spanish, DURATION: 9 months, EXPERIENCE: 0 to 3 years, and PLACES AVAILABLE: 30.

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CUNEF UNDERGRADUATE CUNEF GRADUATE CUNEF EXECUTIVE

Master in Data Science for Finance

PROGRAM: Postgraduate courses MODE: Full-time LANGUAGE: Spanish DURATION: 9 months EXPERIENCE: 0 to 3 years PLACES AVAILABLE: 30

Hospital example (III)

- **Predictive analytics** may **forecast** a surge in patients admitted to the ER in the next several weeks.
- For example, we can use the input data to predict a range of new patients that may be presented to us according to different scenarios. We can get data from other hospitals and incorporate it into our algorithm.



Hospital example (III)

- Predictive analysis allows us to obtain probable scenarios given parameters on infections, measures taken, etc.
- For example, how different arrival rates affect waiting time, bed and staff occupancy, care for other patients, etc., with an estimate of which is more likely.



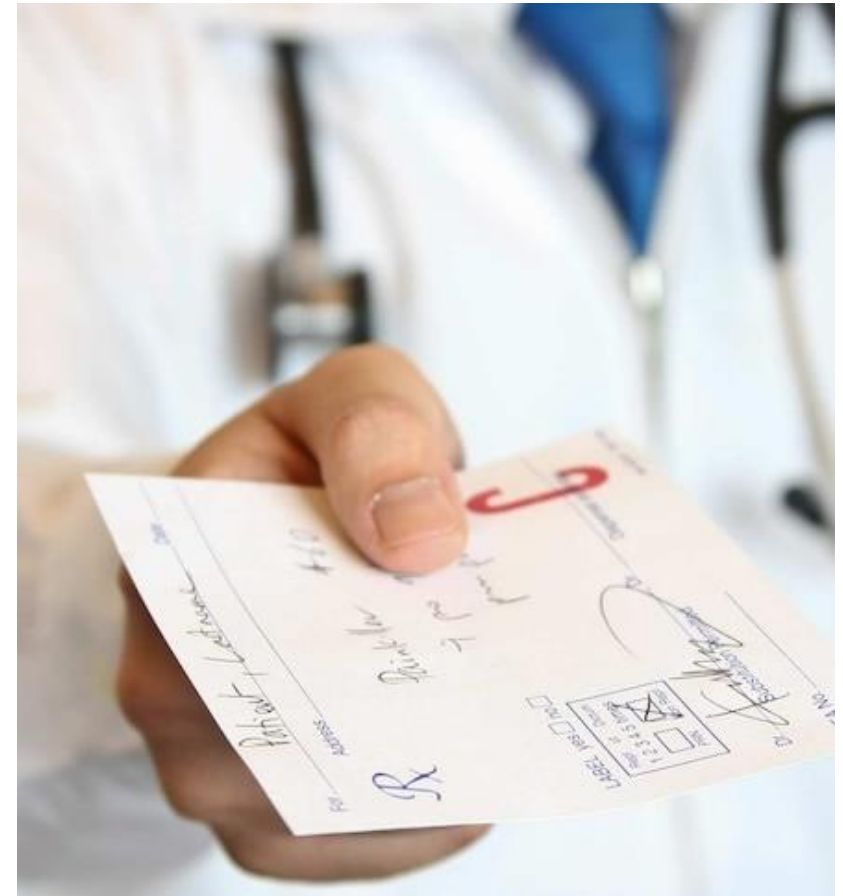
Types of Data Analytics Table of Contents

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Prescriptive Analytics

- The next **step up** in terms of **value** and **complexity** is the prescriptive model.
- “**How to Make it Happen**”
- Deliver business value through **recommendations**.



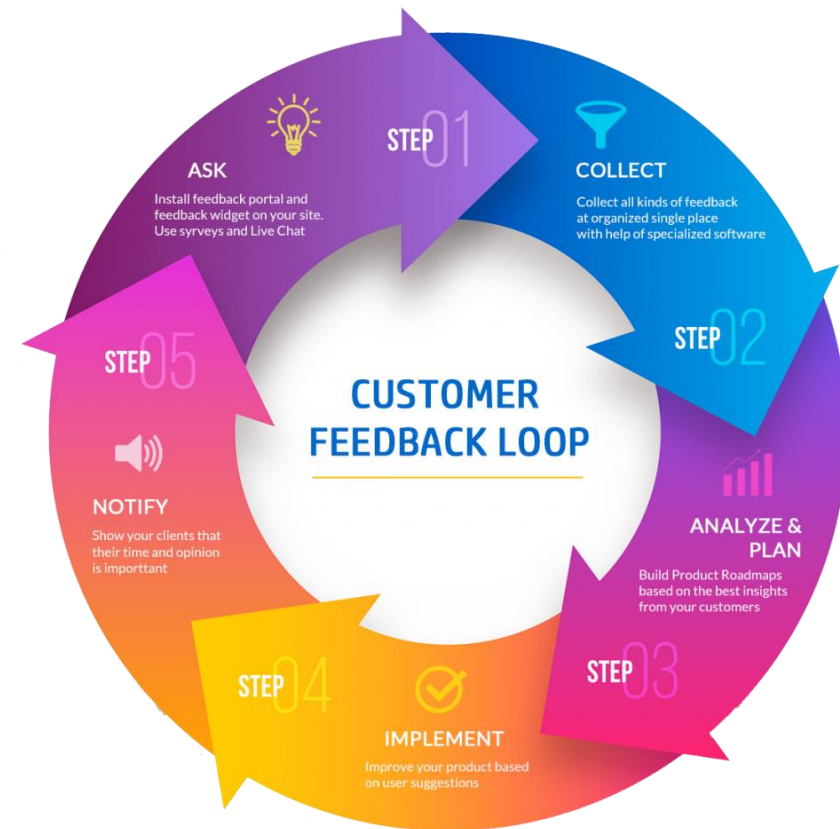
Prescriptive Analytics

- Prescriptive analytics builds on predictive analytics by **helping determine recommended** (prescribed) **actions** based on desired potential (predicted) outcomes, helping organizations **achieve** their **business objectives**.



Prescriptive Analytics

- Prescriptive analytics models are **constantly “learning”** through **feedback mechanisms** to continuously analyze action and event relationships and recommend the **optimal solution**.



Prescriptive Analytics

- By **simulating** the solution, prescriptive analysis can examine all key performance criteria to ensure that, **before anything is implemented**, the outcome would achieve the objectives set or respect certain operational limits.



Prescriptive Analytics

- Artificial intelligence, Machine Learning and Neural Network algorithms are often employed to support prescriptive analytics by helping to make specific suggestions **based on nuanced patterns and perceptions of organizational goals, limitations and influencing factors.**



Hospital example (IV)

- An advanced prescriptive analysis system can automatically recommend measures to mitigate the effect of the new pathogen, for example:
 1. Suggesting extra safety measures for workers and patients at risk.
 2. Recommendations on staffing needs, whether to hire more or not, and with what profile.
 3. Whether or not additional equipment is needed, where to buy it and at what price.



More Examples of each type of Data Analysis

Improving Sales Performance with Data Analytics

- 1. Descriptive Analytics (What happened?)**
 - Analyze past sales data (e.g., "Sales grew 10% in Q3, dropped 5% in Q4").
- 2. Diagnostic Analytics (Why did it happen?)**
 - Identify causes (e.g., Q3 growth due to promotions; Q4 decline from stock shortages).
- 3. Predictive Analytics (What will happen?)**
 - Forecast future trends (e.g., "15% sales growth predicted in Q2 based on rising demand").
- 4. Prescriptive Analytics (What should we do?)**
 - Recommend actions (e.g., "Increase inventory for Product B by 20% before Q2").



More Examples of each type of Data Analysis

Improving User Engagement for a Streaming Media Service with Data Analytics

1. Descriptive Analytics (What happened?)

- Analyzes past user behavior (e.g., "User viewership increased 25% in the last quarter, with peak activity on weekends").

2. Diagnostic Analytics (Why did it happen?)

- Identifies reasons for trends (e.g., "Increase driven by new content releases and targeted marketing campaigns").

3. Predictive Analytics (What will happen?)

- Forecasts future behavior (e.g., "Predict a 30% increase in streaming during the holiday season based on past patterns").

4. Prescriptive Analytics (What should we do?)

- Recommends actions (e.g., "Release original content ahead of holidays and offer personalized recommendations to boost engagement").



More Examples of each type of Data Analysis

Netflix and Data Analytics: Account Sharing Policy

1. Descriptive Analytics (What happened?)

- Analyzed usage data and identified that 30% of accounts were shared across multiple households.

2. Diagnostic Analytics (Why did it happen?)

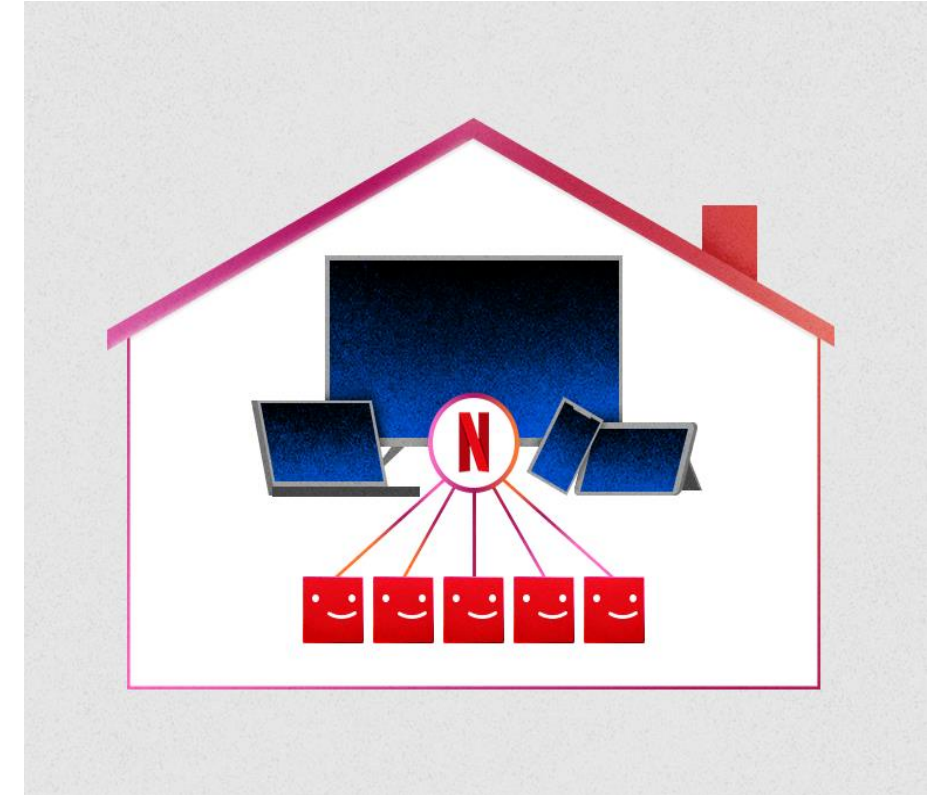
- Investigated user behavior and found that shared accounts contributed to lower subscription growth rates.

3. Predictive Analytics (What will happen?)

- Forecasted potential revenue increases by enforcing stricter account sharing policies, predicting a rise in new subscriptions.

4. Prescriptive Analytics (What should we do?)

- Recommended implementing measures to limit account sharing, such as requiring verification for new devices to promote individual subscriptions.



Let's train our own Neural Network

Teachable Machine



More videos at <https://www.youtube.com/playlist?list=PLJfHZtseuscuTQfodmFnbZ3rBgCWsRT9t>

Takeaways for the Topic

Types of Data Analytics

- The 4 types of Data Analytics.
- Both descriptive analytics and diagnostic analytics look to the past to explain what happened and why it happened.
- Predictive analytics and prescriptive analytics use historical data to forecast what will happen in the future and what actions you can take to affect those outcomes.
- Forward-thinking organizations use a variety of analytics together to make smart decisions that help the business.



Skills you acquired in this topic

- How to build your own Neural Network using Teachable Machine.

